SHRADDHA PRASHANT MANDHARE

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PROFESSIONAL SUMMARY

Detail-oriented Biomedical Engineering graduate student (GPA 3.94) with hands-on experience in automation system design, validation, and prototype development in regulated environments. Proven ability to drive process efficiency improvements and cross-functional collaboration. Seeking a Manufacturing Engineer role in the medical device industry to apply expertise in SolidWorks-based design, process validation, GMP compliance, and Lean/Six Sigma principles to deliver safe, high-quality products.

CORE COMPETENCIES

Design & CAD: SolidWorks, Fusion360, AutoCAD, CAD modeling, DFM/DFA principles **Process & Quality:** GMP, ISO 13485, Lean Six Sigma (DMAIC), CAPA, 5S, Root Cause Analysis

Validation & Testing: IQ/OQ/PQ, FAT/SAT, Process FMEA, DHF, ISA-88, ISO 9001 **Regulatory:** FDA 21 CFR Part 820, 510(k) pathway, ISO 10993, IEC 60601, Traceability

Engineering Tools: MATLAB, R, Microsoft Office Suite, HMI/SCADA, PLC/DCS programming, DeltaV

EDUCATION

M.S. Biomedical Engineering – University of Florida *Gainesville, FL | Aug 2025 – Aug 2027 | GPA: 3.94*

B.Tech. Instrumentation & Control Engineering – Cummins College of Engineering *Pune, India | Aug 2020 – Jun 2024 | GPA: 3.43*

PROFESSIONAL EXPERIENCE

Graduate Engineer Trainee *Nashik, India · Jun 2024 – Jun 2025*

Emerson Export Engineering Centre II

- Designed and validated 30+ HMI graphics for pharmaceutical batch processes using **ISA-88 modular architecture**; ensured compliant, **GMP-ready** batch execution across manufacturing lines.
- Led end-to-end design and commissioning of two safety automation subsystems, managing interlock logic and fail-safe testing achieving a **15% improvement in testing efficiency** and faster commissioning timelines, equivalent to Lean cycle-time reduction principles.
- Executed requirements traceability and functional testing against ISA-88, GMP, and client standards; maintained documentation packages supporting regulatory compliance and design history file (DHF) practices.
- Applied structured **root-cause troubleshooting during FAT**, identifying and resolving systemic configuration defects directly reducing rework and production delays.

Engineering Intern *Nashik, India · Jul 2023 - Dec 2023*

Emerson Export Engineering Centre II

- Executed Pre-Factory Acceptance Testing (Pre-FAT) and Factory Acceptance Testing (FAT) for 20+ Distributed Control System (DCS) cabinets across oil & gas projects, verifying hardware integrity, I/O wiring, control logic, communication protocols, and system functionality prior to site deployment.
- Collaborated closely with clients and cross-functional engineering teams during acceptance testing to identify, troubleshoot, and resolve 20+ configuration and integration issues, reducing rework, minimizing commissioning delays, and improving on-time project delivery.
- Ensured all testing, documentation, and execution activities complied with **ISO 9001 quality management standards**, maintaining traceability, adherence to project specifications, and consistent delivery of quality-assured automation systems.

RESEARCH EXPERIENCE

Graduate Research Assistant *Gainesville, FL · Mar 2025 – Present*

Neural Interface Technology Lab – University of Florida

- Developing a multi-layer EEG head phantom using ballistic gelatin as a brain-equivalent medium with embedded dipole antennas to characterize an implantable PCB's signal acquisition applying biomaterial selection, prototype assembly, and pre-clinical bench testing methodology.
- Conducting IACUC-compliant animal research for ferritin drug studies; responsible for experimental setup, controlled data collection, and quantitative analysis.

PROJECTS

Wearable Vacuum-Assisted Urinary Collection Device (GatorCatch)

- Contributed to prototype design and assembly using SolidWorks-modeled components; evaluated silicone, TPU, PDMS, and latex-free elastomers against biocompatibility and manufacturability criteria applying DFM/DFA principles and mapping ISO 10993 & IEC 60601 standards to upstream design requirements.
- Researched FDA 510(k) dual-predicate pathway for Class II clearance; conducted IP prior art analysis across USPTO, EPO, and WIPO; developed commercialization plan targeting the $4.2B global urinary incontinence device market.

Multimodal Emotion Recognition – Wearable Sensor System

- Designed a wearable signal acquisition system integrating EMG, PPG, and pulse pressure sensors; extracted 15+ features including HRV, Spectral Flux, and MAV, then deployed an attention-based deep multimodal architecture to classify emotional states with improved generalization.
- Executed full-cycle development from sensor selection and signal conditioning through feature extraction and output verification directly applicable to medical device data acquisition and V&V testing workflows.

Customer Discovery & Technology Commercialization (i-Corps–Style)

- Conducted 15 structured customer discovery interviews with clinicians, researchers, and industry stakeholders using NSF i-Corps Business Model Canvas methodology to evaluate commercial viability of faculty-assigned biomedical IP.
- Identified unmet clinical needs, product-market fit, and regulatory/manufacturing scale-up barriers; synthesized findings into a go/no-go commercial assessment targeting the $4.2B global medical device market.